db.js

db.js — SQLite Schema and Utilities

- Purpose: Initializes the SQLite DB, defines schema, and provides helper functions and seed data.
- DB: `better-sqlite3` with WAL mode and foreign keys.

Tables

- `users`: accounts for clients and officers; includes unique_code, account_number, balance, status, last_active.
- `sessions`: persisted hashed JWT tokens with expiry timestamps.
- `friends`: social connections with `pending/accepted/rejected`.
- `transactions`: transfer history; includes fraud flags and score.
- `fraud_alerts`: alerts generated by fraud module; joined with users and transactions.
- `notifications`: user notifications (txn events, friend events, system).
- `login_attempts`: brute-force tracking.

Helpers

- Generators: `generateUniqueCode`, `generateAccountNumber`, `generateTxnId`, `generateAlertId`.
- User getters: `getUserById`, `getUserByEmail`, `getUserByCode`, `getUserByAccount`.
- `safeUser(user)`: strips sensitive fields (e.g., `password_hash`).
- Uniqueness: `getUniqueCode`, `getUniqueAccountNumber` loop until unused.
- `seedOfficer()`: creates default officer user on first run.

Impact on System

- Provides strong boundaries for persistence and lookup used by auth, transactions, social, and fraud modules.